

Prior versus Residual

Homeostatic and allostatic control laws for agent memory updates

Richard Emate

richard@theemate.com · Independent · 16 August 2026

Preprint. Empirical companion to Emate (2026), doi:10.5281/zenodo.21962419. Results from VoltMem 0.4.0. Not a theory of consciousness.

Abstract

Agent memory systems that scale overwrite protection by a domain volatility prior V_d treat two different quantities as one number: a slow belief about how often a *kind* of fact changes, and a fast residual about whether *this* observation was unexpected. We separate them. A **homeostatic** law charges V_d in both the evidence score and the threshold. An **allostatic** law drops V_d from the score and scales the threshold by leftover surprise $r_t = |M_t - \hat{E}_t| / \sigma_t$, the distance from predicted mismatch rather than from stored text. A **composite** gate uses the allostatic law only for an explicit high-mismatch correction or an unexpected residual against a learned $\hat{E}$; otherwise it keeps the homeostatic insurance.

On scripted probes with oracle domain and mismatch, dropping V_d from the score recovers explicit recency-shift (entrenched career change) as a cliff at exponent $p=0$, not a blend. The same drop produces 20% more false updates under the classifier's real error structure, because the double V_d charge was insurance against a mislabeled stable trait plus weak evidence. Composite matches homeostatic's false-update rate (94.9%, 0.93 false updates / 18) while keeping the recency-shift win. Defining surprise as leftover after anticipation makes a predicted weak stream go quiet; catching that stream is a sleeptime job on time-decayed belief mass, not a live EMA of raw mismatch. Sixteen daily weak mentions supersede overnight; the same sixteen monthly do not.

The overwrite law is only reached after a match. Through *remember(text)*, decision error once routed is about six points; similarity and linking dominate. Topic similarity is non-separable for must-link versus must-not-link pairs. Two-stage recall-then-verify with a conservative local model takes irreversible errors to zero on a combined update+coexist harness. We do not claim a public-benchmark win. We claim a measured decomposition: prior and residual are different jobs; a switch beats a blend; linking sits in front of both laws.

1. Introduction

A memory layer for LLM agents has to decide, on each write, whether a new sentence updates a stored fact or sits beside it. Facts do not share a timescale. Personality traits should resist a single offhand comment. Locations should move. Jobs sit in between: they change, but not every Tuesday.

VoltMem [1] scales protection by a domain volatility prior $V_d \in (0, 1]$. Low V_d raises the bar for overwrite; high V_d lowers it. The same idea, applied to Elastic Weight Consolidation [2], is a causal control knob on the stability–plasticity tradeoff, not a free-lunch accuracy win: shuffling or inverting the domain→volatility map must degrade performance monotonically (REAL > SHUFFLE > SWAP). That preprint treats V_d as one sufficient statistic for how readily a memory should move.

It is not. A channel can be historically volatile and currently well-predicted, or historically stable and currently in a regime change. One scalar cannot see the difference. Continual-learning instinct says volatile weights should stay plastic. Predictive-processing instinct says a residual of expected size is not a reason to reopen a belief [3, 4]. Both uses of V_d are valid. They must not share a multiplier.

This paper tests that split on the VoltMem write path.

Contributions.

- We show that the homeostatic law *double-charges* V_d (numerator of E_t and denominator of θ_t), and that this is load-bearing for two opposite failures: it blocks explicit career change after entrenchment, and it insures mislabeled stable facts against weak evidence.
- We define online surprise as leftover mismatch after anticipation, $r_t = |M_t - \hat{E}_t| / \sigma_t$, with V_d widening σ . An EMA of raw M_t is not surprise; it sits on a knife-edge under a constant weak stream.
- We show that blending the V_d exponent between 0 and 1 inherits neither win. A discrete **composite** gate does.
- We restore accumulated weak evidence as a sleeptime detector on time-decayed belief mass. Spacing, not count, is the diagnostic.
- We measure that the control law is a small term in end-to-end *remember()* error, and that topic similarity cannot separate same-fact updates from coexisting facts. Two-stage linking, not a threshold, is the architectural repair.

We do not claim that these control laws implement consciousness, allostasis in Sterling's physiological sense [8], or a new continual-learning algorithm on Split-MNIST. The four-way EWC comparison outlined in the research log was not run. The evidence here is scripted agent-memory probes.

2. Related work

Uniform and volatility-weighted consolidation. EWC [2] applies one global elastic penalty, anchored to old weights. Emate [1] scales that penalty per domain by V_d measured before the update. The present paper leaves retrieval and EWC aside and asks only when a *symbolic* memory slot should overwrite.

Surprise-gated plasticity. Gating learning on prediction error is not new. Liakoni et al. [5] accumulate reward-rate mismatch across timescales and raise synaptic plasticity when unexpected uncertainty exceeds expected uncertainty. SuRe [6] ranks replay by negative log-likelihood. Farhang et al. [7] use a predictor over a frozen encoder both to gate episodic writes and as a metacognitive signal. Our r_t is the same family of idea, applied to a memory *slot* rather than a synapse or a replay buffer: surprise is distance from predicted mismatch, not from stored text, and not from a lifetime counter.

Expected versus unexpected uncertainty. Yu and Dayan [4] distinguish noise the system should already have budgeted for from a change in the world's contingencies. Domain V_d is our expected-uncertainty prior. Residual r_t is unexpectedness at a step. Collapsing them is the defect we measure.

Agent memory stores. Mem0 [10], Zep/Graphiti, and related systems retrieve by embedding and then add, update, or skip. We do not compare product quality. We measure a sequential structure they share with VoltMem: matching happens before the overwrite rule. A miss never consults the control law.

3. Control laws

On *observe()*, an extractor supplies mismatch magnitude $M_t \in [0, 1]$, source reliability R_t , and (optionally) a domain. Let C be confirmation count, G_t a goal-delta factor, L_t load (here 1). Residual evidence is

$$\text{res}_t = (M_t R_t / C^\alpha) G_t, \quad \alpha = 0.6$$

Homeostatic (the original VoltMem law [1]):

$$E_t = \text{res}_t \cdot V_d, \quad \theta_t = \theta_0 / V_d \cdot L_t$$

Update iff $E_t > \theta_t$. V_d shrinks the score and raises the bar. For a medium-stable job ($V_d \approx 0.3$), evidence is cut to a third while the bar more than triples.

Allostatic:

$$E_t = \text{res}_t, \quad \theta_t = \theta_0 / V_{\text{trait}} \cdot L_t \cdot s(m)$$

where V_{trait} is the domain prior (not a drifted EMA), and $s(m) \in [S_{\text{min}}, 1]$ is a decreasing function of recent leftover surprise. V_d sets the bar once.

Residual surprise. Let $\hat{E}$ and σ be a per-item running mean and scale of mismatch, with σ widened by V_d so ordinary noise on a volatile channel is expected:

$$r_t = \min(1, |M_t - \hat{E}_t| / (\sigma_t \cdot Z)), \quad Z = 3$$

E_t still uses M_t . Surprise uses r_t . Confirms pull $\hat{E}$ down. Time decay on a persisted *surprise_at* (30-day half-life) is the route back to settled. A lifetime *mismatch_count* has no such route; it only ratchets open.

Composite is a switch, not an exponent $p \in (0, 1)$:

- allostatic if $M_t \geq 0.85$ and the source is an explicit statement, or if $\hat{E}$ has been learned and $r_t \geq 0.5$;
- otherwise homeostatic.

A fresh item has no $\hat{E}$, so the first weak blip cannot open the easy-update path.

Sleeptime. Online r_t asks whether this *step* was unexpected. After a few similar asides, $\hat{E} \approx M$ and $r_t \rightarrow 0$. Accumulated weak evidence is scored later as time-decayed mass $\sum M_t R_t^{1/2^{\text{age}/30d}}$ against a bar $0.35 / V_d$, ignoring rows before the last confirm. Lifetime counts are not read.

4. Experimental setup

Probes are scripted. Unless noted, *observe()* is called with *domain=* and *mismatch_magnitude=*, so Batteries A–E isolate the control law from routing. Negative control on A: REAL > flat > swap of the V_d map. Errors on E and J are typed: a **false update** or **false merge** destroys a stored fact (irreversible); a missed update or duplicate leaves both facts retrievable.

Battery	What it asks	Path
A	Retain/update labels under real / flat / swap priors	<i>observe(domain, M)</i>
C	Explicit recency-shift (career change after quiet; preference control)	same
D	Weak slow-burn (sixteen casual mentions; daily vs monthly)	same
E	Classifier label noise at the real confusion structure, and at 50% mislabel	same
F/J	End-to-end <i>remember(text)</i> : update + coexist	matcher + law
G/H	Must-link vs must-not-link pairs, held-out 56	linking only

The heuristic classifier is ~84% accurate on a 230-utterance corpus. Ground truth on E stays tied to the true domain: a mislabel does not change whether the memory ought to update.

5. Results

5.1 Two ingredients, two jobs

Sweeping the V_d exponent in E_t ($p=1$ homeostatic, $p=0$ allostatic) crossed with $s(m)$ on/off: only $p=0$ recovers recency-shift, and it is a cliff. At $p=0$ the entrenched career change clears the bar by ~27%; at $p=0.25$ it misses by ~6%. $s(m)$ changed no outcome at any p on Battery C, because every C probe ends in explicit $M=0.90$, which clears a medium-band θ -cap regardless of surprise.

Battery A remains 20/20 under real priors with REAL > flat > swap intact.

5.2 The double charge is insurance

At the classifier's real error rate:

law	accuracy	false updates / 18
homeostatic	94.9%	0.93
allostatic	93.8%	1.12

composite	94.9%	0.93
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Allostatic loses 1.1 points and produces 20% more false updates. Under 50% mislabel the gap widens (3.44 vs 2.84 false updates). Every allostatic-only failure has the same shape: a very-stable fact (*personality_trait* $V=0.05$, *biographical* $V=0.10$) misread into a more volatile band, then contradicted by weak evidence. Removing V_d from E_t fixes career changes and breaks mislabeled traits. It is a trade. Partial p buys nothing (cliff at 0). Composite is the remaining option: it matches homeostatic on E and allostatic on C.

5.3 Surprise is leftover, not raw difference

An EMA of raw M_t under a constant weak stream drove θ from 0.500 to 0.2941 against $E_t = 0.2940$ — $2e-4$ above the trigger. Nine of ten $S_{\min} \times$ half-life settings caught the change; the shipped 14-day half-life was the failing corner. That is calibration of a quantity that is not surprise.

After $r_t = |M_t - \hat{E}_t| / \sigma_t$:

battery	allostatic
C recency-shift	hold (still rides dropping V_d from E_t)
D daily weak stream	never — after a few hits $\hat{E} \approx M$, $r_t \rightarrow 0$
E label noise	unchanged (fresh item, $s(m)$ unread)

Same words after two weeks of asides are not surprising. That is the definition working. Catching the pile is not a live-EMA job.

5.4 Composite and sleeptime

battery	composite
A real priors	20/20
C recency-shift	hold (career U, preference R)
D live weak stream	never
E real label noise	94.9% / 0.93 FU (identical to homeostatic)
E 50% mislabel	84.2% / 2.84 FU (identical to homeostatic)

Sleeptime on the logged pile:

stream	live	overnight consolidate()
16 daily weeks, <i>professional_context</i>	never	supersedes
same 16, monthly	never	does not rewrite
16 daily weeks, <i>core_preference</i>	never	does not rewrite

Identical evidence, identical count, only spacing differs. A counter that never decays cannot see this. Time-decayed belief mass can.

5.5 The law is behind the matcher

remember(text) only, 18 update probes:

similarity	labels	routed	correct	correct routed
keyword	oracle	22.2%	22.2%	100%
hashing	oracle	27.8%	27.8%	100%
sentence-transformers	oracle	77.8%	72.2%	92.9%
sentence-transformers	shipped	55.6%	44.4%	80.0%

Error budget: similarity ≈ 50 points; classifier-via-routing ≈ 28 ; residual routing ≈ 22 ; decision once routed ≈ 5.6 . Batteries A–E cannot see the first three. Two of four probes that never route even with embeddings are the explicit career-change and goal-change cases allostatic was built to recover. End-to-end allostatic vs homeostatic at the best configuration is 77.8% vs 72.2% — one probe in 18, not significant.

5.6 No threshold separates topic from identity

Held-out 56 pairs (28 must-link, 28 must-not-link). Ranking is inverted: “proficient in Python” vs “proficient in Japanese” scores 0.80 on the default scorer; the career change that must link scores 0.25. Non-separability replicates on keyword, hashing, and sentence-transformers. Embeddings double must-link recall (12/28 → 25/28) at almost no change in false-merge *count* (13/28 → 12/28), and raise *severity*: keyword false merges mostly discard the incoming fact; embedding false merges supersede the stored one.

Stage-1 recall at bar 0.20 is 27/28 held-out. A perfect verifier therefore scores 55/56. A cheap lexical verifier (cardinality + change marker) is 19/24 on the fitted split and 35/56 held-out — worse than the embedding ladder (41/56). An LLM verifier asking “same subject?” and “same attribute?” (attribute = the question a fact answers, never the answer) scores 52/56 hosted (*gpt-4o-mini*) and 49/56 local (*qwen2.5-coder:14b*) with **0 false merges**. The hosted model is three pairs higher and two irreversible losses worse. Under the paper's error taxonomy the local model is preferred. Prompt framing moved the hosted model from 29/56 to 52/56; the local model moved 48 → 49 and never collapsed on the badly posed prompt. A 3B rubber-stamp (24 false merges) and a 14B instruct model that almost always says KEEP_BOTH (5/28 must-link) show that model choice does not track “bigger is better.”

On the combined 18 update + 14 coexist harness:

configuration	update	coexist	overall	irrev.
keyword, shipped labels	22.2%	64.3%	40.6%	5
keyword, oracle labels	22.2%	35.7%	28.1%	9
embeddings, shipped	44.4%	78.6%	59.4%	4
embeddings, oracle	72.2%	71.4%	71.9%	4
embeddings + verify, either labels	55.6%	100%	75.0%	0

Under the threshold ladder, improving the classifier makes data loss worse: correct labels put two distinct facts in the same slot. Two-stage linking removes that incentive (coexist 100% either label condition). Allostatic vs homeostatic on this combined set is 78.1% vs 75.0%, zero irreversible errors either way — still one-probe scale.

6. Discussion

The prior and the residual are different jobs. Charging V_d twice is not a bug in general: it is the wrong setting for an explicit correction of an entrenched medium-stable fact, and the right setting for weak evidence on a misfiled trait. A blend does not interpolate those cells. A gate does.

Online surprise must not be an average of contradiction. If it is, a predicted stream impersonates a regime change until a half-life is nursed across a line. Leftover-after-anticipation habituates, which is what the definition asked for, and it therefore cannot catch a slow pile. Horizon belongs to a decayed accumulator, preferably off the interactive path.

None of that matters if the new sentence never finds the slot. The control-law debate is about six points in an error budget whose first fifty are the similarity function. We report that not as a reason to abandon the law, but as a reason not to over-read end-to-end allostatic-vs-homeostatic tables.

7. Limitations

- Probes are synthetic and small. Battery F/J rows other than the 50- and 28-point gaps should not be read individually.
- Recency-shift still sits on a hand-tuned θ -cap (*EXPLICIT_E_RATIO*). Robustness of $p=0$ to domain misclassification was the original open risk; composite addresses the measured cell (weak evidence on a settled item) rather than proving robustness in general.
- The Split-MNIST four-variant design (EWC vs V_d vs residual vs residual+mode-switch, with forgetting-after-recency as the key diagnostic) was not executed.

- Related work on surprise-gated *weight* plasticity is active; we did not re-run those methods on our probes. The claim is about a symbolic memory slot.
- Linking results depend on a 24/56 split and on prompt wording for the hosted verifier. Held-out pairs were written from a structural grid after the first 24; they are not a public benchmark.

8. Conclusion

A volatility prior answers how readily a *kind* of memory should move. Leftover surprise answers whether this step was already priced in. Using one number for both produces a characteristic pair of failures, and blending the number does not fix the pair. A switch, a residual that habituates, and a sleeptime accumulator that sees spacing, together implement the split. The matcher still sits in front. Duplicates are recoverable; silent overwrites are not.

References

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Code: github.com/Rouche01/voltmem · Package: *voltmem* 0.4.0 · Companion: <https://doi.org/10.5281/zenodo.21962419>